Courses

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Electrical & Comp. Engineering (ECE)
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Electrical & Comp. Engineering (ECE)

ECE 145 - Introduction to Computer Programming

Description: Specification, design, implementation, testing, debugging, applications in numerical analysis and computer simulations. Control structures, functions, arrays and other data types. Programming in C, Python, and Matlab.

Prerequisite(s): MATH 120 Minimum Grade of C OR MATH 120E Minimum Grade of C.
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Infrequently
Attribute(s): Breadth-Info & Comm in Society

ECE 198 - Electrical and Computer Engineering Work Experience I

Description: Supervised work experience with agency, firm, or organization which uses engineers. Intended for students who have part-time cooperative experience jobs. Limited to students enrolled in more than 6 credit hours. Student must be a declared major in electrical and computer engineering.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Instructor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 199 - Electrical and Computer Engineering Cooperative Education I

Description: Supervised work experience with agency, firm, or organization which uses engineers. First work period of five year academic/work experience program. Requires consent of advisor.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Advisor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 210 - Circuit Analysis I

Description: DC and AC steady-state circuit analysis. Loop and nodal analysis; network theorems; phasors; complex power; and single-phase and three-phase circuits. IAI Course EGR 931.

Prerequisite(s): MATH 150 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 3
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): IM-Engineering

ECE 211 - Circuit Analysis II

Description: Time-domain transient analysis; complex frequency; frequency response; two-port networks; Laplace transform techniques; and impulse response and convolution. Three hours lecture and one laboratory session per week.

Prerequisite(s): ECE 210 Minimum Grade of C AND MATH 250 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 4
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall

ECE 282 - Digital Systems Design

Description: Concepts and design of digital and computer circuitry; binary number systems; study of combinational and sequential logic circuits; introduction to Verilog HDL. Laboratory exercises involve circuit implementation and HDL programming.

Prerequisite(s): CS 140 Minimum Grade of C OR CS 145 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 4
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): IM-Engineering

ECE 298 - Electrical and Computer Engineering Work Experience II

Description: Supervised work experience with agency, firm, or organization which uses engineers. Intended for students who have part-time cooperative experience jobs. Limited to students enrolled in more than 6 credit hours.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Instructor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 299 - ECE Cooperative Education II

Description: Supervised work experience with agency, firm, or organization which uses engineers. Second work period of five year academic/work experience program. Requires consent of advisor.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Advisor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 326 - Electronic Circuits I

Description: Introduction to semiconductors; diode, transistor and FET; small and large signal analysis; and logic gate families and design. Three hours lecture and one laboratory session per week. Student must be a declared major in an engineering discipline.

Prerequisite(s): ECE 211 Minimum Grade of C AND MATH 305 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 4
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 340 - Electromagnetics

Description: Introduction to engineering electromagnetics. Includes vector analysis, time-harmonic fields, electromagnetic wave propagation, transmission lines, waveguides, antennas.

Prerequisite(s): (PHYS 142 Minimum Grade of C OR PHYS 152 Minimum Grade of C) AND PHYS 152L Minimum Grade of C AND ECE 211 Minimum Grade of C AND MATH 305 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 3
Next available term: Spring 2027
Term frequency: Spring; Fall

ECE 341 - Princ w/Electro Mec Enrg Conv

Description: Basic electromagnetic concepts; energy-based torque and force and calculations; transformers; induction machines; synchronous machines; and DC machines. Three hours lecture and one laboratory session per week.

Prerequisite(s): ECE 340 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 4
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 351 - Signals and Systems

Description: Basics of continuous and discrete signals and systems. Convolution; Fourier analysis; filtering; modulation and sampling; and Z-transforms. Student must be a declared major in an engineering discipline.

Prerequisite(s): ECE 211 Minimum Grade of C AND MATH 305 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Majors: Electrical Engineering, Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall

ECE 352 - Probability and Statistics

Description: Probability; random variables and probability distributions; statistics; Monte-Carlo simulations; estimation theory; decision theory; hypothesis testing; random processes; and linear system response to random processes.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 365 - Control Systems

Description: Mechanical and electrical systems modeling, signal flow graphs, state variable approach, root-locus approach, Bode ploys/Nyquist plots, frequency domain design and Proportional-Integral-Derivative (PID) controller turning methods.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Spring; Fall

ECE 375 - Introduction to Communications

Description: Time- and frequency-domain analysis; and bandwidth, distortion, and noise. Baseband pulse transmission; sampling; pulse shaping. Digital and analog modulation techniques. Analysis of bit-error probability.

Prerequisite(s): ECE 351 Minimum Grade of C AND ECE 352 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Summer; Spring; Fall

ECE 381 - Microcontrollers

Description: Interfacing and programming microcontrollers to measure/control various hardware and signals, GPIO, LCD, ISRs, encoders, UART/RS-232, I2C, SPI, ADC/DAC. Three hours lecture and one laboratory session per week.

Prerequisite(s): ECE 282 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 398 - Electrical and Computer Engineering Work Experience III

Description: Supervised work experience with agency, firm, or organization which uses engineers. Intended for students who have part-time cooperative experience jobs. Limited to students enrolled in more than 6 credit hours.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Instructor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 399 - ECE Cooperative Education III

Description: Supervised work experience with agency, firm, or organization which uses engineers. Third work period of five year academic/work experience program. Requires consent of advisor.

Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman; Sophomore, Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Advisor
Credit Hours: 0
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall
Attribute(s): Cooperative Education Course

ECE 404 - Electrical and Computer Engineering Design

Description: Design overview, methodologies, considerations, constraints, scheduling, and management. Focus on both oral and written communication. Students work in teams while rotating leadership responsibilities.

Prerequisite(s): ECE 282 Minimum Grade of C AND ECE 351 Minimum Grade of C AND (ECE 365 Minimum Grade of C OR ECE 375 Minimum Grade of C OR ECE 381 Minimum Grade of C).
Restriction(s): May not be enrolled as the following Classifications: Freshman, 1st Semester; Freshman; Junior; Sophomore, Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 405 - Electrical and Computer Engineering Design Laboratory

Description: Realization of a term-based senior project, including construction, computer simulation, debug, and test as required by the project to obtain a functional prototype.

Prerequisite(s): ECE 404 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 426 - High Frequency Design

Description: High frequency circuit design with elements of RF engineering. Amplifiers, oscillators, modulators, impedance matching, switching, signal integrity, and tuning.

Prerequisite(s): ECE 326 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Fall

ECE 427 - Knowledge-Based Systems

Description: Engineering-oriented perspective on artificial intelligence (AI) technology. General AI concepts and specifically knowledge-based intelligent systems and machine learning applied to engineering problem-solving.

Prerequisite(s): CS 140 Minimum Grade of C OR CS 145 Minimum Grade of C.
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Summer

ECE 428 - Analog Filter Design

Description: Active and passive filter synthesis. Standard low-pass approximations: Butterworth, Chebyshev, Inverse Chebyshev, Cauer, Bessel and frequency transformations. Active and passive circuit implementations.

Prerequisite(s): ECE 351 Minimum Grade of C AND ECE 326 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Fall

ECE 429 - Bioinstrumentation

Description: Design and use of biosignal sensors, bioamplifiers, and filters for measuring physiological data; emphasizes origins and characteristics of nerve and heart signals; includes cell analysis and dialysis machine design.

Prerequisite(s): ECE 327 Minimum Grade of C or admission to graduate Engineering Program (GM).
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Occasionally

ECE 436 - Digital Signal Processing

Description: Discrete-time signals and systems; sampling; z-transforms; discrete Fourier transform; difference equations; design and implementation of digital filters; and DSP development systems.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Spring; Fall

ECE 437 - Medical Imaging Systems

Description: Basic signals and systems, imaging principles, and image quality measures for X-ray radiography, X-ray computed tomography, ultrasound, and magnetic resonance imaging.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Occasionally

ECE 438 - Image Analysis and Computer Vision

Description: Image formation, geometrical and topological properties of binary images; image filtering; boundary detection; image segmentation; and pattern recognition. Two hours lecture and one laboratory session per week.

Prerequisite(s): ECE 351, or ME 356, or consent of instructor
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Fall

ECE 439 - Digital Image Processing

Description: Fundamentals of human perception; sampling and quantization; image transforms; enhancement; and restoration and coding. Two hours lecture and one laboratory session per week.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Spring

ECE 444 - Power Electronics

Description: Basics of DC/DC and DC/AC conversion, inductors, transformers, switching characteristics of semiconductor devices, elements of electromagnetic compatibility.

Prerequisite(s): ECE 326 Minimum Grade of C or admission to graduate Engineering Program.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Spring

ECE 445 - Power Distribution System

Description: Distribution system planning; load characteristics; application of distribution transformers; design of distribution system; voltage-drop and power-loss calculations; voltage regulation; and protection and reliability.

Prerequisite(s): ECE 341 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Spring

ECE 446 - Power System Analysis

Description: Synchronous machines; power transformers; transmission lines; system modeling; load-flow study; economic operation of power systems; symmetrical components; symmetrical and unsymmetrical faults; and power system stability.

Prerequisite(s): ECE 341 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Summer

ECE 447 - Radar Systems

Description: Introduction to radar systems, including antenna fundamentals, radar equation, radar signals and systems, CW radar, FM-CW Radar, pulse radar, and tracking radar.

Prerequisite(s): ECE 340 Minimum Grade of C AND ECE 351 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Spring

ECE 455 - System Modeling & Optimization

Description: Mathematical modeling of engineering systems; dynamic response of electrical and mechanical systems; and optimization models in electrical engineering.

Prerequisite(s): ECE 351 Minimum Grade of C (concurrency allowed)
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Occasionally

ECE 465 - Control Systems Design

Description: Root-locus analysis; frequency-response analysis; design and compensation technique; describing-function analysis of nonlinear control systems; and analysis and design by state-space methods.

Prerequisite(s): ECE 365 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Fall

ECE 466 - Digital Control

Description: Topics include finite difference equations; Z-transforms and state variable representation; analysis and synthesis of linear sampled-data control systems using classical and modern control theory. Same as ME 466.

Prerequisite(s): ME 450 Minimum Grade of C OR ECE 365 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Occasionally

ECE 467 - Robotics-Dynamics and Control

Description: (Same as ME 454 and MRE 454) Robotics; robot kinematics and inverse kinematics; trajectory planning; differential motion and virtual work principle; and dynamics and control.

Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Instructor
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Fall

ECE 476 - Electronic Circuits II

Description: Small signal analysis, transistor amplifier design, frequency response, feedback system analysis, output stage design, signal generation and waveform shaping circuits. Three hour lecture and one hour laboratory session per week.

Prerequisite(s): ECE 326 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Not Currently Scheduled
Term frequency: Spring

ECE 477 - Network Engineering

Description: Principles and practices of network engineering with particular emphasis on the physical, data-link, and network layers as applied to telecommunication and computing systems.

Prerequisite(s): ECE 282 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Spring 2027
Term frequency: Spring; Fall

ECE 482 - Microprocessor System

Description: Design of microprocessor systems using VLSI building blocks. Several microprocessors and peripheral ICS studied laboratory experiments with microprocessor systems using logic analyzers. Three hours lecture and one laboratory session per week.

Prerequisite(s): ECE 282 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 3
Next available term: Fall 2026
Term frequency: Fall

ECE 483 - Advanced Digital Systems Engineering

Description: Design of digital systems using a hardware description language, and logic synthesis tools. To master advanced logic design concepts, logic optimization, including the design of synchronous combinational and sequential circuits using state of the art CAD tools.

Prerequisite(s): ECE 282 Minimum Grade of C.
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 0 to 3
Next available term: Spring 2027
Term frequency: Summer; Spring

ECE 484 - Digital VLSI Design

Description: Discussion of CMOS circuits, MOS transistor theory, CMOS processing technology, circuit characterization, and CMOS Circuit and Logic Design.

Prerequisite(s): ECE 326 Minimum Grade of C
Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Credit Hours: 3
Next available term: Fall 2026
Term frequency: Fall

ECE 491 - Independent Study

Description: Individual investigation of a topic in electrical engineering to be agreed upon with the instructor. May be repeated to a maximum of 6 hours provided no topic is repeated. Requires consent of instructor.

Restriction(s): Must be enrolled in one of the following Majors: Computer Engineering,Computer Engineering,Electrical Engineering
Registration Consent: Instructor
Credit Hours: 1 to 4
Next available term: Not Currently Scheduled
Term frequency: Summer; Spring; Fall

ECE 492 - Topics in Electrical and Computer Engineering

Description: Selected topics of special interest. Course schedule will include name of topic. May be repeated to a maximum of 6 hours so long as no topic is repeated. Requires consent of instructor.

Restriction(s): Must be enrolled in one of the following Colleges: School of Engineering
Registration Consent: Instructor
Credit Hours: 2 to 6
Next available term: Fall 2026
Term frequency: Summer; Spring; Fall